

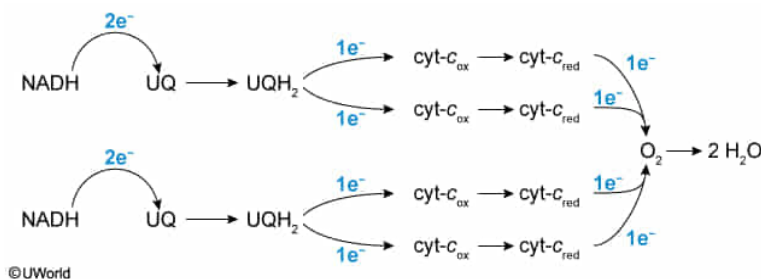
In the electron transport chain, four electrons are required to fully reduce one O_2 molecule to two H_2O molecules. Given this information, what is the minimum number of NADH molecules required to reduce a molecule of oxygen?

- ☐ A. 1 [6%]
✔ ☒ B. 2 [59%]
☐ C. 3 [2%]
☐ D. 4 [31%]

Correct

60% answered correctly

Explanation:



The **electron transport chain (ETC)** is a series of protein complexes in the inner mitochondrial membrane that facilitate transfer of high-energy electrons from reduced energy carriers to lower-energy acceptors. The energy released by this series of redox reactions drives ATP synthesis. **NADH carries two electrons** in the form of a hydride H^- ion, and these electrons enter Complex I of the ETC to reduce ubiquinone (UQ) to ubiquinol (UQH_2).

UQH_2 then travels through the lipid bilayer to Complex III, where the electrons from a single UQH_2 can reduce up to two cytochrome c (cyt c) molecules. These cyt c_{red} molecules then travel through the intermembrane space to Complex IV, where they deposit

electrons one by one to oxygen, progressively reducing it to water once **four electrons** have been deposited. Note that although electrons can also be deposited into the ETC via FADH_2 in Complex II, electrons from NADH alone are sufficient to reduce oxygen to water.

NADH is a **two-electron carrier**. Because turning O_2 into water requires **four electrons** per oxygen molecule, a minimum of **two NADH molecules** would need to pass their electrons through the ETC to fully reduce O_2 .

(Choice A) If NADH could carry four electrons instead of two, only one NADH molecule would be required to fully reduce O_2 to water.

(Choice C) NADH carries two electrons, so two NADH molecules carry the four high-energy electrons required to fully reduce O_2 .

(Choice D) Four single-electron carriers, such as cyt c, would be required for full reduction of O_2 . However, NADH carries two electrons.

Educational objective:

Electron donors and acceptors in the electron transport chain have different carrying capacities. Full reduction of oxygen to water requires four electrons, so two copies of a two-electron carrier such as NADH would be required to fully reduce oxygen. Likewise, four copies of a one-electron carrier, such as reduced cytochrome c, are required.

Time spent:0 sec

Subject:
Biochemistry

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System:
Metabolic Reactions